

Chapter 1

An Overview on Classification of Energy Storage Systems

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The predominant concern in contemporary daily life is energy production and its optimization. Energy storage systems are the best solution for efficiently harnessing and preserving energy for later use. These systems are categorized by their physical attributes. Energy storage systems are essential for reliable and green energy in the future. They help balance the ups and downs of renewable energy sources, like when the sun isn't shining or the wind isn't blowing. This means we can use more clean energy, which is good for the environment. With better technology and recycling, these systems will become even more important. This chapter aims to give readers a clear understanding of the various energy storage systems.

Introduction

The energy demand across diverse applications, from industrial to household utilities, has been experiencing rapid growth. This surge can be attributed to several factors, such as population expansion, the adverse environmental impact of fossil fuels, and the increasing reliance on digital technologies. Consequently, the surplus energy generated from these resources necessitates efficient storage for future utilization. Similarly, the energy produced from renewable energy resources are reliable to meet the energy demand in small scale applications but fails to meet the large-scale industrial applications.

The grid performance of the renewable energy sources were limited due to the following factors such as uncertainty and variability in the power output, system stability and reliability. Integrating Energy storage systems with renewable energy resources overcomes the above issues by acting as either power sources or function as a system (or) device that controls the power fluctuation and improves the power quality. In addition to this, the storage of electrical energy from AC system

directly as an electric current is not reliable. Hence, the conversion of AC electricity to various other forms of energy sources leads to the development of different types of energy storage systems namely electrical energy, chemical energy, thermal energy, electrostatic energy, and electromagnetic energy storage systems. In the following chapters, we will discuss in detail about each energy storage systems. The efficiency performance of the energy storage systems is summarized in Figure 1.

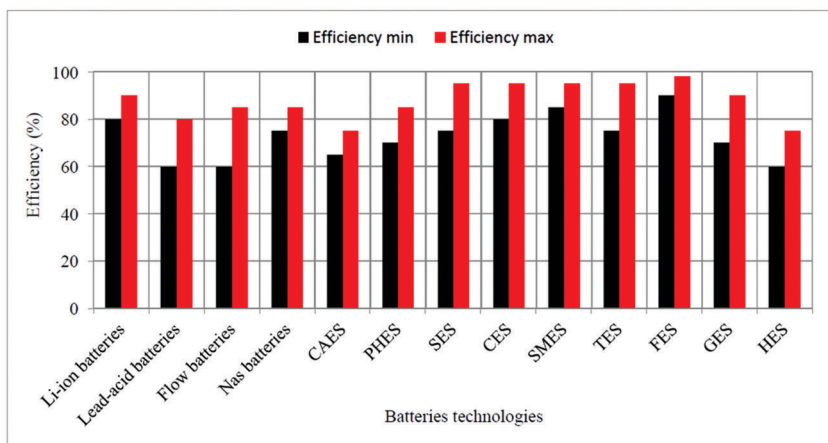


Figure 1. Efficiency of the energy storage systems. Reproduced from reference (1). Copyright The Authors. Distributed under a Creative Commons Attribution License (CC BY).

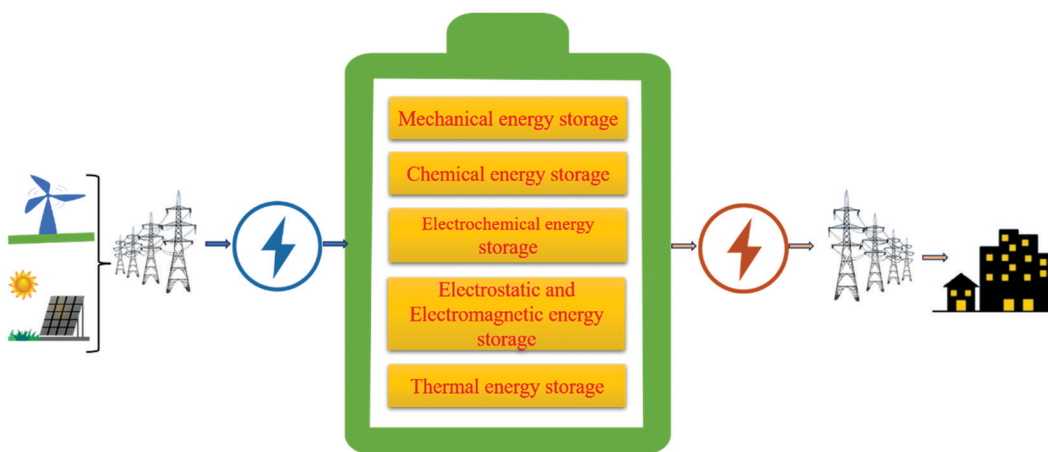


Figure 2. Classification of energy storage systems.

Classification of Energy Storage System

These fundamental energy-based storage systems can be categorized into three primary types: mechanical, electrochemical, and thermal energy storage. Furthermore, energy storage systems can be classified based on several criteria, such as the type of stored energy, the technology employed, their intended application, and their capacity (1, 2). These classifications lead to the division of

energy storage into five main types: i) mechanical energy storage, ii) chemical energy storage, iii) electrochemical energy storage, iv) electrostatic and electromagnetic energy storage, and v) thermal energy storage, as illustrated in Figure 2.

These classifications provide a framework for understanding the diverse ways in which energy can be stored and utilized efficiently. Each type of energy storage has its advantages and limitations, making them suitable for different applications and contexts.

Mechanical Energy Storage

Mechanical energy storage systems are most commonly used throughout the world due to their advantages, which include their capability to quickly convert and release stored mechanical energy. These systems store energy by converting electrical energy into mechanical energy in either potential or kinetic forms. Additionally, they play a vital role in integrating renewable energy sources into the power grid. Unlike fossil fuel power plants which can be easily adjusted to meet demand fluctuations, renewable sources like solar and wind are variable in nature. Mechanical energy storage helps bridge this gap by storing excess energy generated during low-demand periods and releasing it back to stabilize the power grid, reduce peak demand, and provide backup power. Mechanical energies are divided into four types: Pumped hydroelectric energy storage, flywheel energy storage, compressed air energy storage, and gravity energy storage. These are prominent examples of widely employed mechanical energy storage systems in energy storage technology (3).

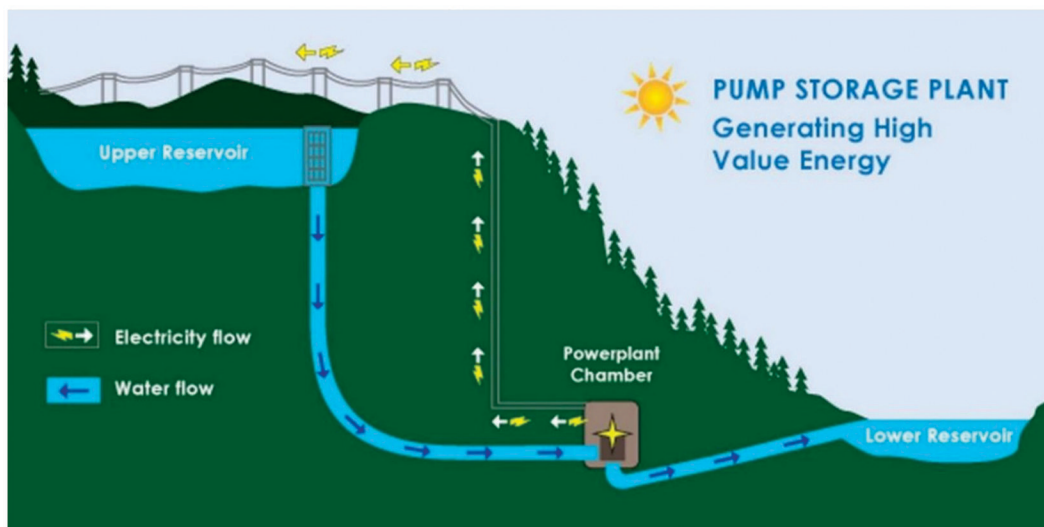


Figure 3. Pumped Hydroelectric energy storage.
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Pumped Hydroelectric Energy Storage (PHES)

The PHES system effectively stores and generates electricity by transferring water between two reservoirs at varying elevations. It is one of the most established and efficient large-scale energy storage and grid stabilization methods. PHS working mechanism involves three levels: i) reservoir (upper and lower reservoir), ii) storage (pumping phase), and iii) energy generating (releasing phase),

shown in Figure 3. i) **Reservoirs:** A pumped hydroelectric storage system consists of an upper and lower reservoir. These reservoirs can be natural or artificial. ii) **Storage (pumping phase):** Excess electricity is used in electric pumps during low electricity demand or high renewable energy generation periods. These pumps move water from the lower reservoir to the upper reservoir. This process effectively stores the excess energy as gravitational potential energy. The water is held at a higher elevation, creating a reservoir of potential energy. iii) **Energy generating (releasing phase):** When electricity demand increases or during periods of low energy generation, the system changes into generating mode. The stored water from the upper reservoir is released back to the lower reservoir, passing through turbines as it descends. The turbines capture this kinetic energy and convert it into mechanical rotation, which the generators transform into electrical energy (4).

This generated electricity is fed into the grid to meet the demand. PHES systems have long lifespans, typically exceeding several decades, with minimal degradation of their energy storage capacity over time, and this system is environmentally friendly (5).

Compressed Air Energy Storage (CAES)

CAES is a technology utilized for storing energy through air compression. This method has found effective application as a large-scale energy storage system, often reaching capacities in the hundreds of megawatts (6). The quantity of energy stored is directly influenced by factors such as the storage container's volume and the specific pressure and temperature conditions under which the air is stored. During periods of excess electricity on the grid, like when the wind is blowing vigorously, or the sun is shining brightly, the extra power is harnessed to operate an air compressor. This compressor pressurizes and stores the air within an underground cavern. When the demand for electricity rises, the compressed air is released from the storage facility and allowed to expand through a turbine. This turbine, in turn, generates electricity that can be seamlessly integrated into the grid (Figure 4) (7). This CAES method was divided into diabatic, adiabatic, and isothermal CAES (8). Most of the CAES plants are currently in the research and development stage.

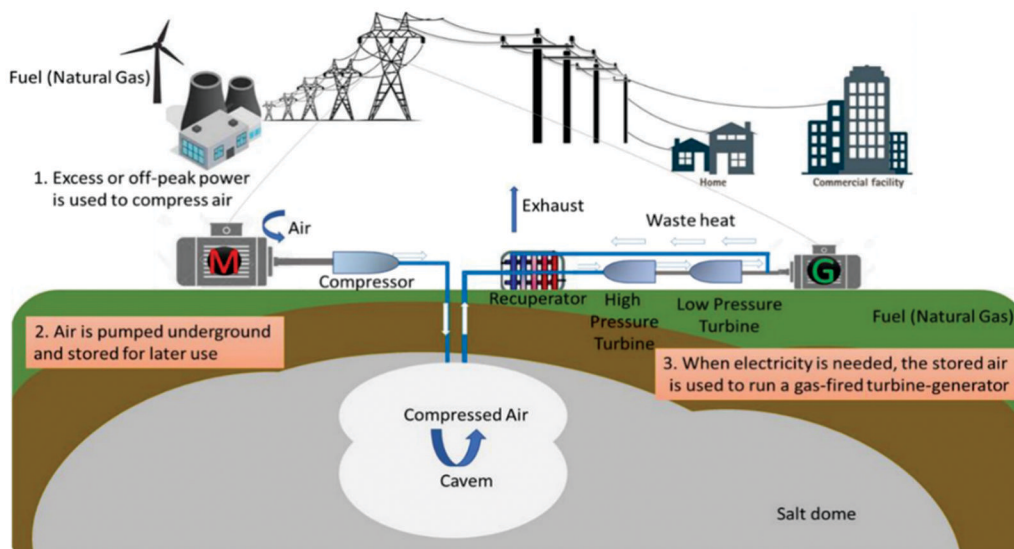


Figure 4. Compressed air energy storage working diagram. Adapted from reference (7). Copyright The Authors. Distributed under a Creative Commons Attribution License 4.0 (CC BY).

Flywheel Energy Storage (FES)

Flywheels are an electromechanical arrangement designed to store kinetic energy by harnessing the angular momentum of rotating masses such as discs or cylinders. Despite their longstanding presence in technology, they have recently gained recognition as a viable solution for storing substantial amounts of energy on a large scale (9). Flywheel energy storage working diagram given in Figure 5.

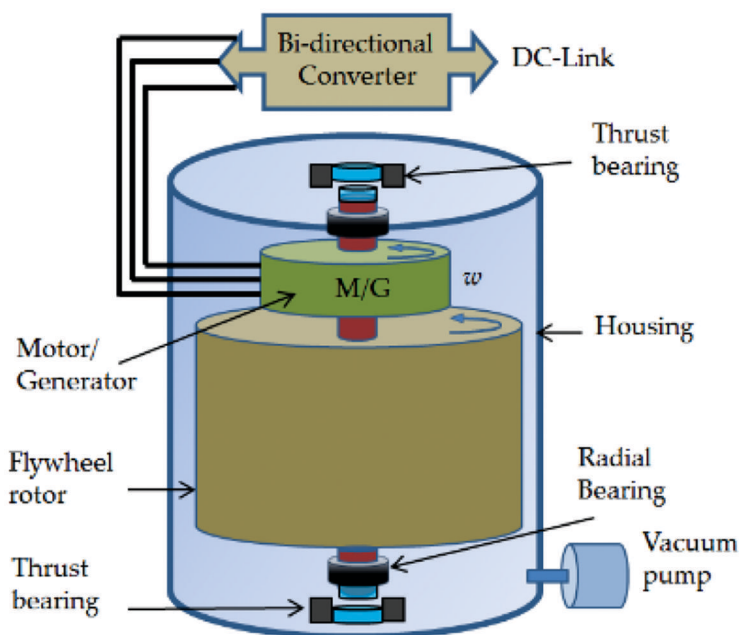


Figure 5. Flywheel energy storage. Adapted from reference (9). Copyright The Authors.

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A flywheel operates by accumulating excess energy within the wheel's rotation (spinning mass) and later converts that stored energy, which is accelerated through the assistance of an electric device functioning as a motor. The spinning wheel acts like a storage bank for energy. This process occurs during the charging of a flywheel energy storage system. The same motor switches to being a generator when it gives out energy from the spinning flywheel to the power system (10). FES systems come in two types: low-speed and high-speed. Low-speed FES is used in industries and spins around 10,000 times per minute or less. High-speed FES is a bit more complex (11). This FES stabilizes power grids, provides backup power, and manages renewable energy fluctuations (12).

Gravity Energy Storage (GES)

Gravity energy storage systems that store and release energy using gravity. These systems work by raising heavy objects to a higher elevation to store potential energy and then allowing them to fall back down to a lower height, changing the potential energy back into kinetic energy and converting it to electricity, as shown in Figure 6 (13). A generating machine lifts a high-weight piston to a higher elevation with abundant power. This work used various methods, such as electric motors, hydraulic pumps, or compressed air. The piston is stored at the higher elevation until it

is needed. When energy is required, the piston is released to a lower elevation, converting the mass's gravitational potential energy into kinetic energy. Then kinetic energy is used to power a generator, which produces electricity. Research into enhancing this technology for future widespread use, encompassing improvements in size, storage capacity, materials utilization, and extended lifespan, should be diligently pursued (14).

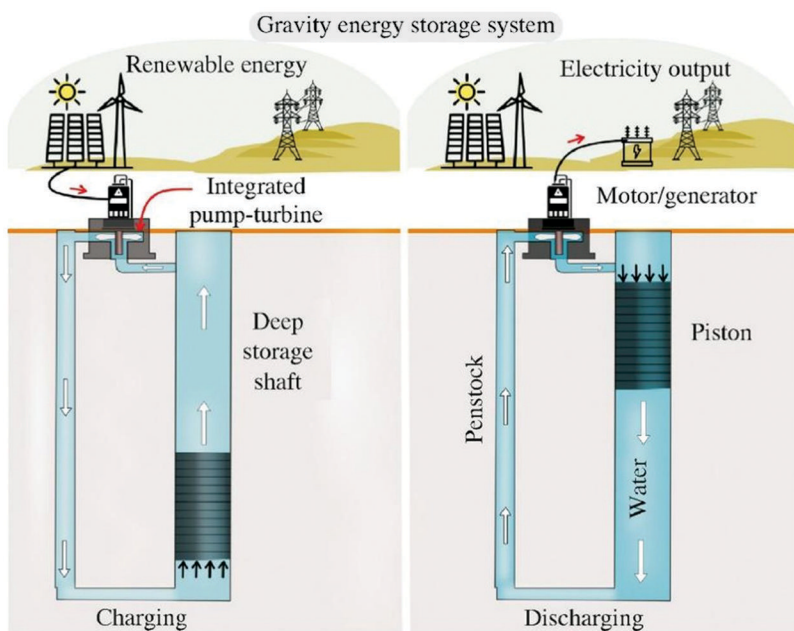


Figure 6. Gravity energy storage. Adapted from reference (13). Copyright The Authors. Distributed under a Creative Commons Attribution License 4.0 (CC BY).

In conclusion, mechanical energy storage systems offer a versatile solution for efficiently storing and releasing energy, playing a crucial role in modern energy infrastructure. They enable the rapid conversion and deployment of stored energy, making them indispensable for grid stabilization and integration of renewable energy sources. Among the prominent types of mechanical energy storage are Pumped Hydroelectric Energy Storage (PHES), Compressed Air Energy Storage (CAES), Flywheel Energy Storage (FES), and Gravity Energy Storage (GES). Each of these systems has its unique working principles and applications, contributing to the sustainability and reliability of our energy networks. As we continue to innovate and improve these technologies, mechanical energy storage will undoubtedly remain a cornerstone of our transition towards a cleaner and more resilient energy future.

Chemical Energy Storage

Chemical energy is stored in the bonds between atoms and molecules. This energy is released when a chemical reaction happens, often transforming the substance into another. We primarily use chemical fuels for making electricity and powering vehicles. Common types of chemical fuels include coal, gasoline, natural gas, and hydrogen. These fuels are turned into heat and mechanical energy, which becomes electricity in power plants and also get electricity directly from stored chemical

energy using fuel cells (it involves electrochemical reactions) (14, 15). Chemical energy storage is a promising technology for storing large amounts of energy for long periods. The most common chemical energy storage systems include hydrogen, synthetic natural gas, and solar fuel storage.

Hydrogen Energy Storage

Hydrogen fuel energy is a clean and abundant renewable fuel that is safe to use. The hydrogen energy can be produced from electrolysis or sunlight through photocatalytic water splitting (16, 17). Hydrogen could become a substantial part of our clean energy solutions with more research (18). Figure 7 shows the working of Hydrogen energy storage system. When there's extra power on the grid, an electrolyzer breaks water into hydrogen and oxygen. Oxygen goes into the air, and hydrogen is stored in a tank, which is stored in various ways, like squeezing it, making it liquid, using special chemicals, and underground storage. When there is demand, then stored hydrogen is converted to electricity through fuel cells. A hydrogen fuel cell is an electrochemical device that converts the chemical energy of hydrogen and oxygen into electricity. Fuel cells are commonly classified based on the type of electrolyte utilized: 1) Proton Exchange Membrane Fuel Cell (PEMFC), 2) Alkaline Fuel Cell (AFC), 3) Phosphoric Acid Fuel Cell (PAFC), 4) Molten Carbonate Fuel Cell, 5) Solid Oxide Fuel Cell (SOFC), and 6) Direct Methanol Fuel Cell (DMFC) (19). The hydrogen fuel cell consists of four main parts: i) anode, ii) cathode, iii) electrolyte and iv) external circuit. i) Anode: This is the negative electrode of the fuel cell. Hydrogen gas is fed to the anode, oxidising it into protons (positively charged hydrogen ions) and electrons. ii) Cathode: This is the positive electrode of the fuel cell, where, Oxygen gas is fed to the cathode, which combines with the protons and electrons to form water. iii) Electrolyte: It allows only protons to flow from the anode to the cathode and restrict the electrons flow through it. iv) External circuit:

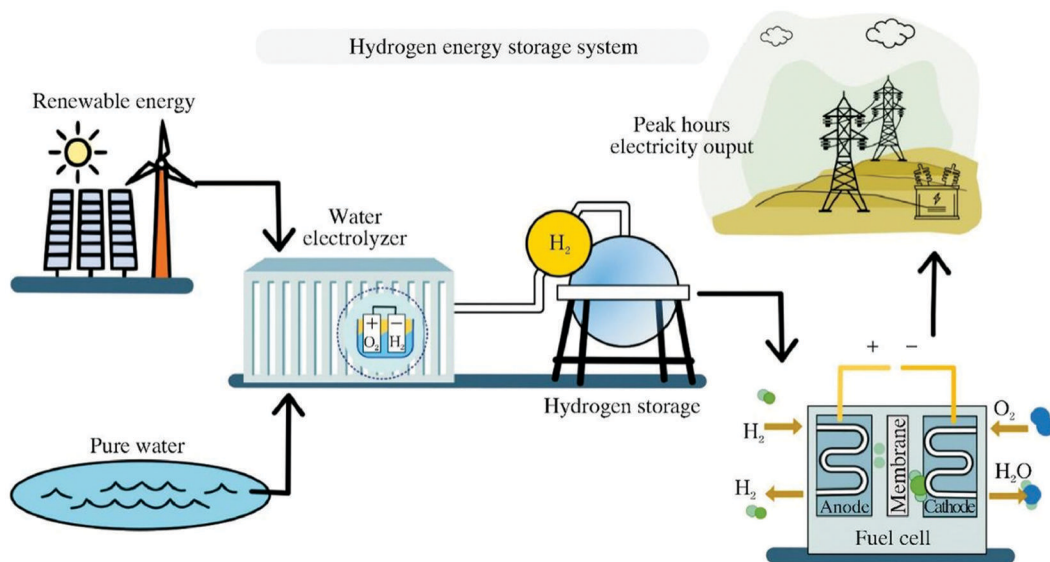


Figure 7. Hydrogen fuel cell energy storage. Adapted from reference (13). Copyright The Authors. Distributed under a Creative Commons Attribution License 4.0 (CC BY).

This is a pathway for the electrons to flow from the anode to the cathode, where, an electric current is produced. The overall reaction that takes place in a hydrogen fuel cell is:



This reaction produces water and heat as the only byproducts. Hydrogen fuel cells are clean and efficient sources of electricity, and hence, they have the potential to be used in a variety of applications, including transportation, power generation, and portable electronics. Hydrogen fuel cells are still under development, but they can potentially be a significant source of clean energy (20, 21).

Synthetic Natural Gas Energy Storage (SNGS)

A SNGS system is a technology that converts electrical energy into synthetic or substitute natural gas, which can be stored and later used as a source of energy (22, 23). The SNGS commonly classified into the following types, namely i) hydrogen, ii) methane hydrate, and iii) biomass SNG. i) Hydrogen-SNG is produced from hydrogen (from renewable sources like solar and wind power) and carbon dioxide (from industrial processes), which are combined to produce methane through a process called methanation. ii) Methane hydrate SNG, a solid methane trapped in water molecules found in marine sediments and permafrost. They were extracted and converted into methane through depressurization process. iii) Biomass-SNG is generated from organic materials like wood, waste, and food scraps through anaerobic digestion, which turns biomass into methane without using oxygen. This methane is collected and stored for energy use.

Solar Fuel Storage

Solar-to-fuel technologies encompass diverse processes designed for harnessing renewable solar energy, converting it into readily usable forms, storing it within synthetic fuels chemical structures, and deploying it as a flexible and sustainable energy source as required (13). This solar fuel storage is done by artificial photosynthesis and thermochemical reactions. 1. Artificial Photosynthesis: Storing solar energy within chemical bonds paves the way for creating an artificial photosynthetic system. In this innovative setup, sunlight is utilized to oxidize water into oxygen, and the resulting electrons can be directed toward two essential pathways: one involving the reduction of carbon dioxide to produce fuels such as methane, methanol, formaldehyde, formate, carbon monoxide, or oxalate, while the other requires proton reduction to generate hydrogen as a clean and sustainable energy resource (24, 25). 2. Thermochemical Reaction: It works at high-temperature thermochemical reactions, such as solar water splitting, to produce hydrogen or carbon dioxide reduction to generate synthetic gasoline at temperatures ranging from 800 to 1,500 °C (26). Solar fuels are still in the early stages of development, but significant research and development are underway to improve the efficiency and cost-effectiveness of solar fuel production. If solar fuels can be produced at a competitive cost, they have the potential to play a significant role in the future of energy (27).

In conclusion, chemical energy storage presents a promising avenue for storing large quantities of energy over extended periods. This storage method utilizes the energy stored in chemical bonds, which can be released through chemical reactions. Among the most common chemical energy storage systems are hydrogen, synthetic natural gas (SNG), and solar fuel storage. As research and development continue to advance these chemical energy storage technologies, they hold significant promise in facilitating the transition towards a cleaner, more sustainable energy future. With further

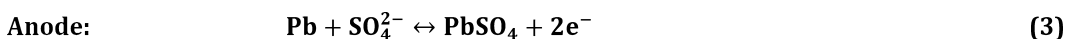
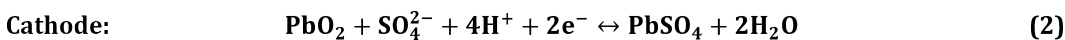
innovation and refinement, chemical energy storage systems could play a crucial role in addressing energy storage challenges and supporting the widespread adoption of renewable energy sources.

Electrochemical Energy Storage

Electrochemical energy storage system undergoes chemical process to store and produce electricity. Batteries are the most widely used electrochemical energy storage systems in industrial and household applications (28). They are classified into two types namely primary and secondary batteries. Among them, primary batteries are the non-rechargeable when the chemicals present in them were completely consumed, whereas, secondary batteries can be recharged and discharged multiple times (29, 30). The rechargeable secondary batteries were widely used in the power system applications. It consists of a cathode, an anode and an electrolyte medium. When the battery is connected to an external circuit, chemical reaction occurs, results in the flow of electrons between the electrodes through the electrolyte and external circuit, and the electromotive force (EMF) is induced (31). The output voltage produced by these batteries will be less than 2 V. To achieve the required voltage and current, they will be connected together in parallel or series configurations. Based on the electrode materials and electrolytes used in the system, the secondary batteries were further classified as Lead-acid battery, Nickel-cadmium battery, Sodium-sulfur battery, Lithium-ion battery and flow batteries (32).

Lead-Acid (LA) Battery

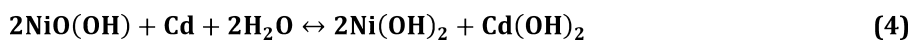
Lead-acid (LA) battery is one of commonly used batteries and the oldest technology developed in 1859. It primarily consists of an anode, a cathode and an electrolyte as a spongy metallic lead, lead dioxide and 37% sulfuric acid (5.99 Molar) (5). The following equations represent the reaction that takes place at the cathode and anode (33).



Both the electrodes were turned into lead sulfate at their discharged state and get back to their original state during charging. This battery technology is heavier and bulkier size and requires more time during charging and discharging. LA batteries can be operated at the temperature between and can deliver the voltage of 2 V. They exhibit 1000 to 2000 cycle life and at low and high temperatures, it shows slow performances. The LA batteries were used in UPS and some spinning reserve applications. They have relatively low power density as 180 W/kg and specific energy as 30 Wh/kg and its infrequent requirement of water maintenance became its major limitation (28).

Nickel-Cadmium (Ni-Cd) Battery

Nickel-cadmium batteries were comprised of an anode made up of cadmium hydroxide, nickel hydroxide as cathode and potassium hydroxide as an electrolyte. The following equation represent the reaction takes place during charging and discharging



While discharging, at positive electrode, Ni(OH)_2 act as active materials. Whereas, in negative electrode, the active material is Cd(OH)_2 . On the other hand, NiO(OH) acts as active material

at positive electrode during charging and **Cd metal** at negative electrode. These batteries need less maintenance and can undergo faster charge/discharge process due to their smaller internal resistance and exhibits long cycle life (3000- 4000 cycles) and energy density as 50-75 Wh/kg. However, due to the toxicity of cadmium, it was replaced by nickel metal hydride battery (Ni-MH). It was widely used in portable devices, UPS, emergency lighting, generator starting and telecoms (34).

Sodium-Sulfur Battery

Sodium-sulfur (NaS) batteries were fabricated using the liquid sulfur as a cathode, liquid sodium as an anode separated by ceramic electrolyte (solid beta alumina) (35).



The above-mentioned equation represents the reversible reaction that takes place in the NaS battery during charging and discharging. The Na⁺ ions will react with the sulfur by passing through the ceramic electrolyte and forms sodium polysulphides (Na₂S₄), during discharge process. The reverse reaction takes place by releasing the Na⁺ from Na₂S₄, that penetrate through the electrolyte, during the charging process. Since, the above reaction is an exothermic reaction and produces heat upto 300°C – 350°C (36). The higher operating temperature and an exposure of sulfur to air and moisture lead to explosion were the major concerns for application purpose. The NaS batteries exhibit 150-240 Wh/kg and 150-230W/kg as an energy and power density with ~2500 cycle life (5).

Lithium-Ion (Li-ion) Battery

Lithium-ion batteries were commercialized in 1991, becomes the most widely used batteries of all types. These batteries (Figure 8) were comprised with lithium metal as the positive electrode (cathode), layered structure of graphite carbon as the negative electrode (anode) and an organic carbonate dissolved with lithium salt (LiPF₆) were used as an electrolyte (35, 37).

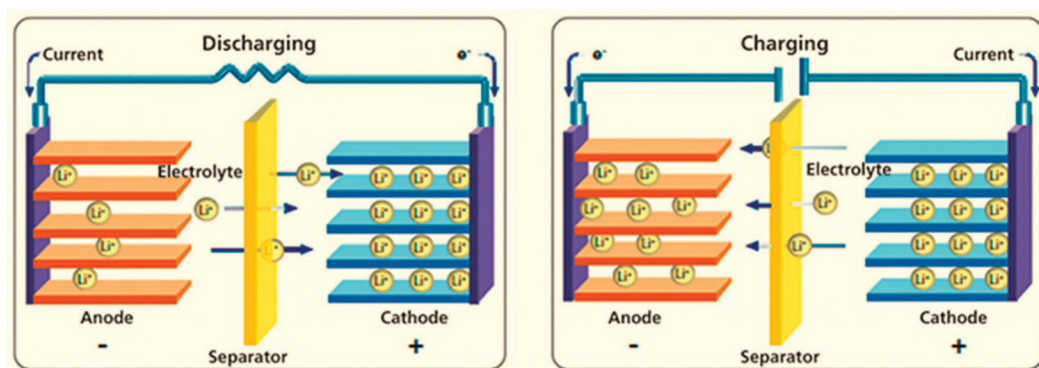


Figure 8. Shows the schematic representation of Li-ion battery. Adapted from reference (37).
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When charging this battery, lithium ions travel through the electrolyte to the negative electrode from the positive electrode and the electrons will travel in the same direction through the external circuit. This process will take place reversibly while discharging the battery. Li-ion battery can deliver

power density upto 1000 - 2000 W/kg and energy density as 100 - 200 Wh/kg and the cell voltage of 3.6 V. The fast charge/discharge cycle, light weight and higher efficiency upto 90% made it as widely applicable battery from smaller electronic devices to large scale applications like car batteries. In spite of their wider applicability, instability at higher temperatures is the problem behind. To overcome this, solid-state Li-ion batteries comprised of solid electrodes and electrolytes that can bear the above-mentioned issue and provides higher energy density (34).

Flow Batteries

Flow batteries have different kind of energy storage properties compared to that of conventional batteries. Flow batteries stores electrical energy externally by forming the oxidized and reduced form of electroactive species in the liquid electrolytes inside the tanks (38, 39).

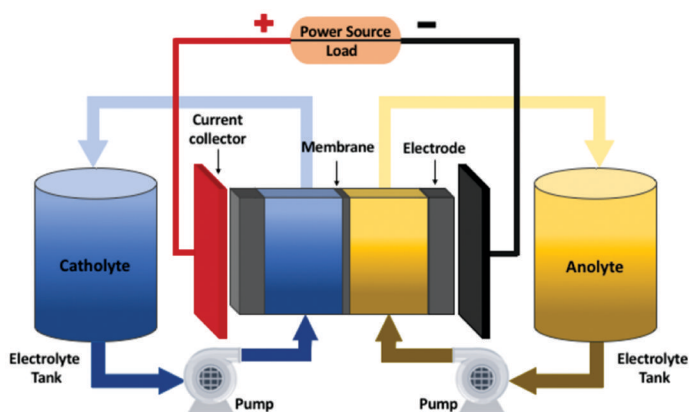
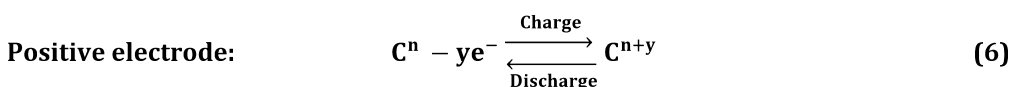
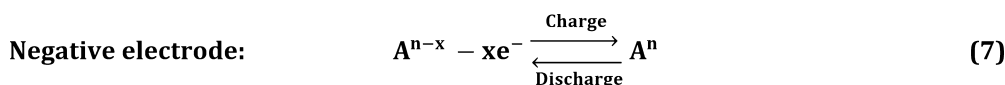


Figure 9. Representation of redox flow batteries. Adapted from reference (39). Copyright The Authors. Distributed under a Creative Commons Attribution License 4.0 (CC BY).

When the electroactive species dissolved in the electrolyte is pumped through the electrochemical cells, it converts chemical energy into electrical energy. Figure 9 shows the redox flow battery system. The energy density of the flow batteries depends on the tank size and the design of the electrochemical cells determines its power density. The process of charge storage is reversible and hence the flow batteries can act like secondary batteries. Flow batteries are classified as redox (reduction-oxidation) flow battery and hybrid flow battery (40). When the electroactive species were dissolved in a liquid electrolyte (i.e., in both the electrodes), it is called as redox flow battery. However, hybrid flow batteries consist of the configurations as that of redox flow batteries except one of the electrodes (either cathode or anode) involving phase changes (solid or gaseous). Zinc/Cerium redox flow battery, Soluble Lead-Acid flow battery and Iron-chromium redox flow batteries were belongs to the types of hybrid redox flow batteries.

Redox flow batteries comprised of two different electrolytes. Among them, one of the electrolyte stores electroactive material for the positive electrode reaction and that of other for the negative electrode reaction (34). The equation 6 and 7 shows the reaction that takes place at the two electrodes.





There are different types of redox flow batteries were available for commercial purposes, namely Vanadium redox flow battery, Bromide/polysulfide flow battery, Zinc/cerium redox flow battery, Soluble Lead-acid flow battery and Iron-chromium redox flow battery (41).

Vanadium Redox Flow Battery (VRB)

The chemical form of energy is stored in VRB by using four different oxidation states of vanadium ions. Namely, V^{2+}/V^{3+} ions couple as negative half-cell electrolyte and V^{4+}/V^{5+} ions as positive half-cell electrolyte and were separated by proton exchange membranes. When charging the battery, the conversion of V^{3+} ions to V^{2+} takes place at negative electrode with the potential of -0.26V vs. SHE (Standard Hydrogen Electrode) and simultaneously, at the positive electrode, the conversion of V^{4+} to V^{5+} occurs at the potential of $+1.00\text{V}$ vs. SHE and the energy was stored as chemical energy in the cell (Figure 10) (42, 43).

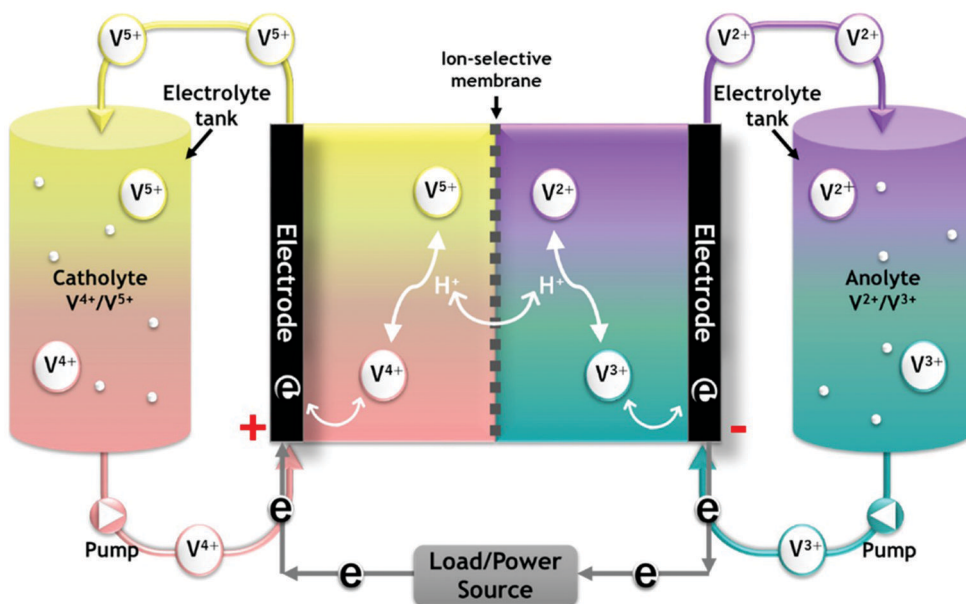
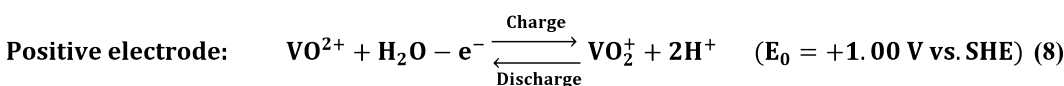
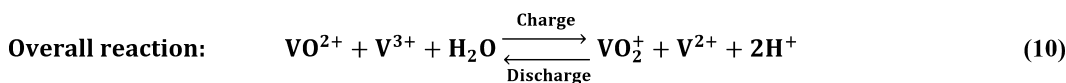
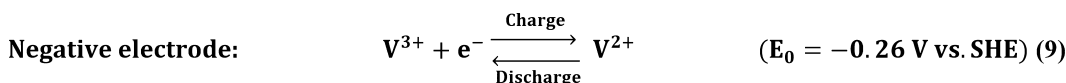


Figure 10. Structure of vanadium redox flow batteries. Adapted from reference (43). Copyright The Authors. Distributed under a Creative Commons Attribution License 4.0 (CC BY).

While discharging, reverse process takes place and the chemical energy that is stored is converted into electrical energy (41, 44, 45). The following equations 8-10 show the reaction that take place during charging and discharging processes (44, 46).

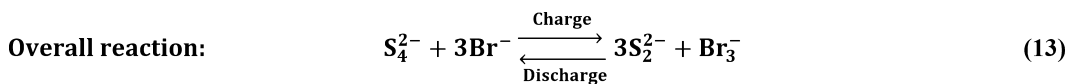
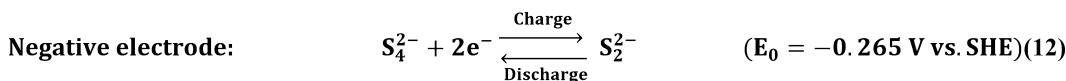
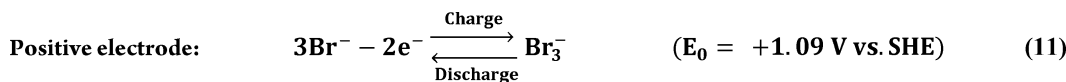




The VRB system exhibits 75-150 W/kg power density, 35-60 Wh/kg energy density and long lifespan of more than 15000 cycles (15-20 years) (41).

Bromide/Polysulfide Flow Battery (PSB)

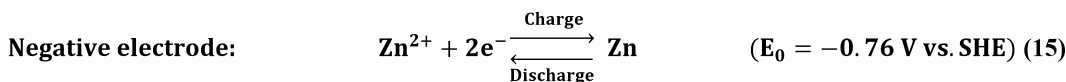
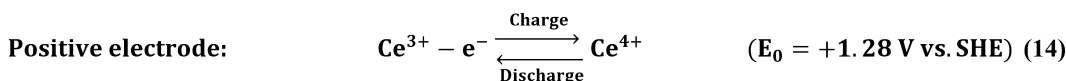
PSB flow battery consists of salt solutions of sodium bromide and sodium polysulfide as electrolytes, separated by membrane that allows positive Na^+ ions for maintaining the electroneutrality of the cell. The equation 11-13 shows the reaction involving at both the electrodes (46).

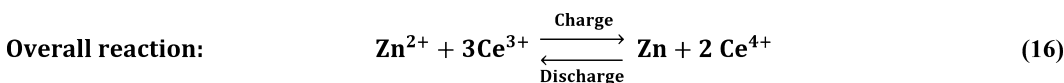


When PSB is charged, at positive electrode, oxidation of three bromide ions takes place and forms tribromide ions. At the same time in negative electrode, sodium particles dissolved in the polysulfide electrolyte were reduced to sulfide ions. On discharging, the same reactions take place in reversible manner. The PSB can be operatable at the temperature of and exhibit a lifespan of 15000 - 20000 cycles. Inspite of their highlights, the PSB performance will be affected by the usage of different electrolyte solutions that causes contamination and formation of crystals of sodium sulfate and bromine will cause environmental issues (41).

Zinc/Cerium Redox Flow Battery

The Zn-Ce redox flow battery is the combination of negative zinc couple and positive cerium couple as electrolytes. This system is under development since 1990s. In 2004, the modified system of this redox flow battery with platinized titanium mesh as cathode and carbon plastic as an anode were developed and were separated by the Nafion membrane. The following expressions (14-16) indicates the reactions at both the electrodes (47),





These oxidation-reduction reactions occur within the respective electrolytes of the Zinc/Cerium Redox Flow Battery, separated by a membrane that allows the transfer of ions while preventing mixing of the electrolytes. This flow battery design enables the storage and release of electrical energy through the reversible conversion of chemical species at the electrodes.

Soluble Lead-Acid Flow Battery

The soluble Lead-acid flow battery (Figure 11) made up of the electrolyte containing single active species (Pb^{2+}). The equations 17-19 show the charge and discharges processes in cathode and anode. On charging this battery, the Pb^{2+} ions were converted from soluble phase to solid phases (Pb and PbO_2). In the discharge process, dissolution takes place. The electrolyte used in this batteries is methane sulphonic acid (48, 49).

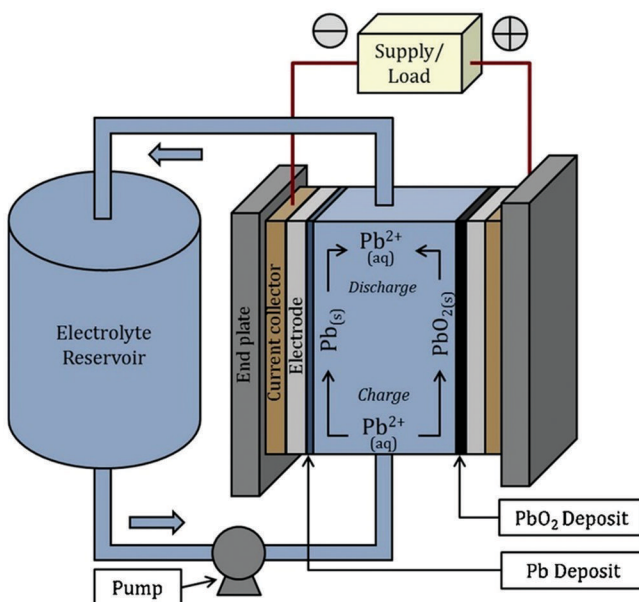
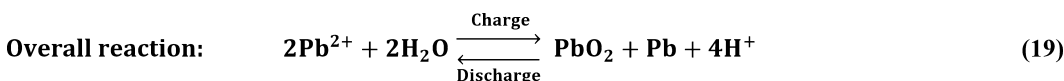
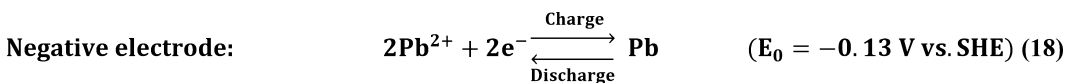
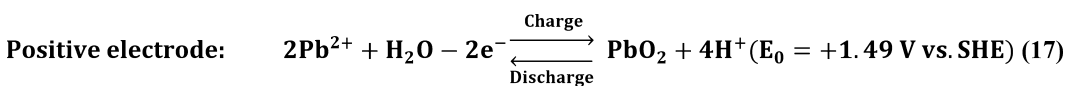
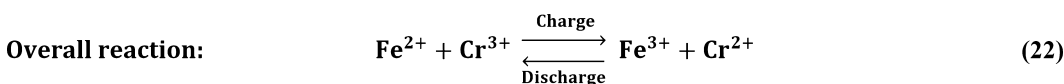
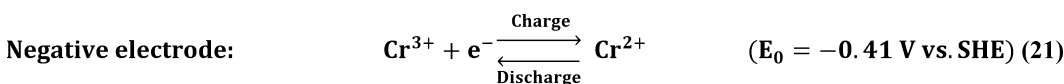
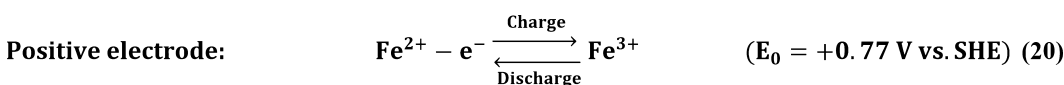


Figure 11. Mechanism of soluble lead acid flow battery. Adapted with permission from reference (48).
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The electrolyte used in this battery typically consists of methane sulfonic acid. During charging, lead ions are converted from the soluble phase to solid phases (lead and lead dioxide), while in the discharge process, dissolution of these solids occurs back into the electrolyte. This reversible conversion of lead species enables the storage and release of electrical energy in the Soluble Lead-Acid Flow Battery

Iron-Chromium Redox Flow Battery

Iron-chromium redox flow battery uses redox couples namely, ferric-ferrous couple dissolved in aqueous solution as positive reactant and negative reactant as chromous-chromic couple and both were acidified with HCl (50). These two half cells were separated by the ion-exchange membrane for the conduction of ions. The following equations were the reactions that take place during charging and discharging process in this system (51).



The last three flow batteries that are discussed above comes under the type of hybrid flow batteries. Since, one of their electrodes involves phase changes (i.e., deposition of ions on the electrode) during the storage processes (52). These systems, represented prominently by batteries, are classified into primary and secondary types, with secondary batteries offering the advantage of rechargeability. Among secondary batteries, a diverse range of technologies exists, each with its unique characteristics and applications. These diverse electrochemical energy storage systems cater to various needs across industries, from small electronic devices to grid-scale applications, with ongoing research aimed at improving efficiency, lifespan, and environmental sustainability.

Electrostatic and Electromagnetic Energy Storage System

Electrostatic and electromagnetic energy storage systems store electrical energy, with no conversion to other forms of energy (i.e., stores as electric field). Capacitors, Supercapacitors and Superconducting magnetic Energy Storage (SMES) belong to this type of energy storage system (32). Figure 12 shows mechanism involved in capacitor and supercapacitor (53).

Capacitor

Capacitors are the energy storage system that stores electric energy in the form of electric field. It consists of non-conducting layer namely dielectric acts as separator, separating the two metal plates as oppositely charged electrodes. It stores direct current at the electrode and dielectric interface by the polarization of dipole around them. The capacitors can accept and deliver high currents but their energy density was very low and its extreme short span of time exhibiting it. It was mainly used in electrical circuits, sensors etc (2, 54).

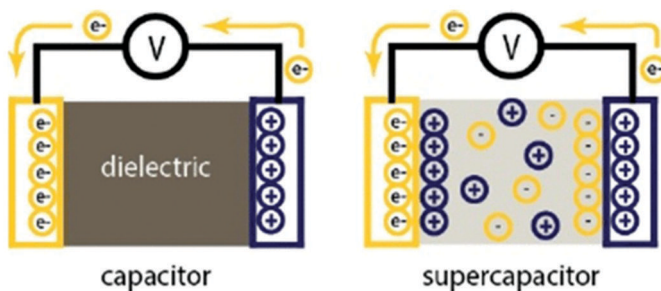


Figure 12. Working mechanism of capacitor and supercapacitor. Adapted from reference (53). Copyright The Authors. Distributed under a Creative Commons Attribution License 4.0 (CC BY).

Supercapacitor

Supercapacitor is also known as ultra-capacitors or double-layer capacitors or electrochemical capacitors (36, 55–57). The supercapacitors are similar to that of conventional capacitors, but, the difference is that the supercapacitor having liquid or polymer electrolytes instead of dielectric medium that are present in the capacitors. Supercapacitors can store 10-100 times more energy than conventional capacitors and response 10 times faster than that of batteries. The charge stored at the interfaces of electrode and electrolyte solution is categorized as double layer capacitors. Whereas, in electrochemical capacitors, the charge store takes place by redox behavior of the metal oxide/sulphides present in the electrode materials (58–60). Since, the supercapacitors doesn't involve any chemical reactions like in batteries, they can able to deliver more power density than batteries. Also, the porous nature and higher surface area of the electrodes helps in more amount of charge storage. Supercapacitor was classified into three types, namely, Electrical double layer capacitor (EDLC), pseudocapacitor and hybrid capacitors. In EDLC capacitors, carbon-based electrode materials were employed (non-faradaic interaction occurs), whereas, in pseudocapacitors, metal oxides, sulphides, selenides, conducting polymers were employed (faradaic interaction occurs) (61). On combining the EDLC and pseudocapacitive materials were used as an electrode material, then it was classified as hybrid capacitors.

$$C \propto \frac{\epsilon A}{d} \quad (23)$$

Equation (23) represents the capacitance (**C**) of a system, where **A** is the surface area of the electrodes, **d** is the separation between the electrodes, and **ε** is the dielectric constant of the separator. This equation highlights the key factors influencing the capacitance of the system, including the size of the electrodes, the distance between them, and the properties of the separator material. The energy density (i.e., the amount of energy stored) of the supercapacitor can be calculated using the formula,

$$E = \frac{1}{2} CV^2 \quad (24)$$

The formula 24 calculates the energy density (**E**) of a supercapacitor, with **C** representing the capacitance value and **V** denoting the applied voltage. Supercapacitors are renowned for their extensive lifespan, boasting cycles ranging from 10^5 to 10^7 , and their rapid response times. This advantageous combination makes them prevalent in various applications, including actuators, engine starters, and electric vehicles, where they excel in regenerating braking systems.

Superconducting Magnetic Energy Storage (SMES)

Superconducting magnetic energy storage (Figure 13) stores electric energy by magnetic field. These magnets were cooled cryogenically below to their superconducting critical temperature (62, 63). Electricity will reach the coil via power conditioning system (i.e., AC to DC converter) and it is stored as magnetic field. On discharging, the current stored in the coil runs to the grid and gets converted by DC to AC converter. The wires will be maintained under superconducting temperature to deplete ohmic losses. The DC current (I) flowing through the coil and the induction of the superconducting coil (L) decides the energy density of SMES (10). It is expressed in the following equation as,

$$E = \frac{1}{2} LI^2 \quad (25)$$

High temperature SMES coil can be operated at and at the temperature of, the low temperature SMES can be operated. SMES exhibits upto 4kW/l of energy density, faster response time and long lifetime of almost 30 years. It was used in the small-scale applications such as UPS (Uninterruptible Power Supply), (Flexible AC Transmission) FACTS and pulse power source. The high cost of superconducting wire and the energy required for maintaining the cooling systems limits it from large application (2, 64).

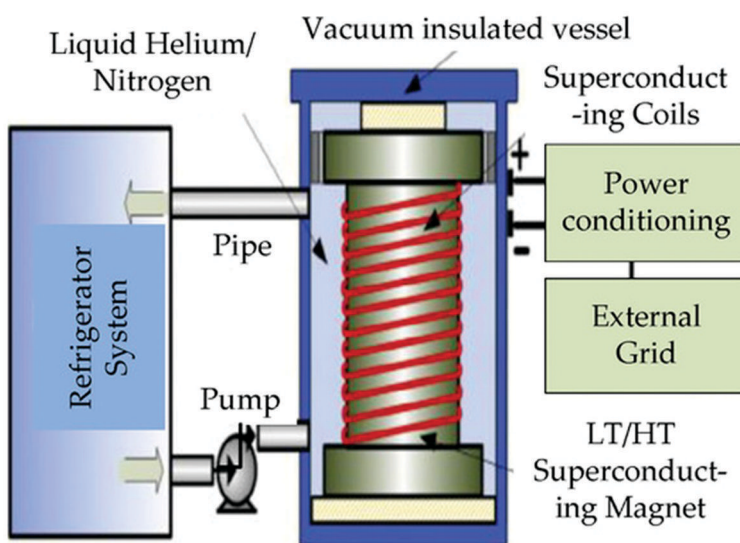


Figure 13. Layout of SME system. Adapted from reference (63). Copyright The Authors. Distributed under a Creative Commons Attribution License 4.0 (CC BY).

These electrostatic and electromagnetic energy storage systems offer vital means of directly storing electrical energy as electric fields. These systems include capacitors, supercapacitors, and Superconducting Magnetic Energy Storage (SMES). Capacitors, characterized by dielectric separators and oppositely charged electrodes, store direct current through dipole polarization, although they have low energy density and short discharge durations. Supercapacitors, on the other hand, boast liquid or polymer electrolytes and can store energy 10-100 times more efficiently than conventional capacitors, with response times 10 times faster than batteries. SMES systems operate

by storing electric energy in a magnetic field generated by superconducting coils cooled below their critical temperature. Despite offering high energy density, rapid response times, and long lifetimes of up to 30 years, SMES systems face challenges such as the high cost of superconducting wire and the energy required for cooling systems, limiting their deployment to small-scale applications.

Thermal Energy Storage (TES)

Thermal energy storage system converts heat energy into electrical energy and stores electricity. It was classified into three types, such as sensible heat, latent heat and thermochemical heat storage system (absorption and adsorption system) (65). Figure 14 shows the schematic representation of each thermal energy storage systems (66).

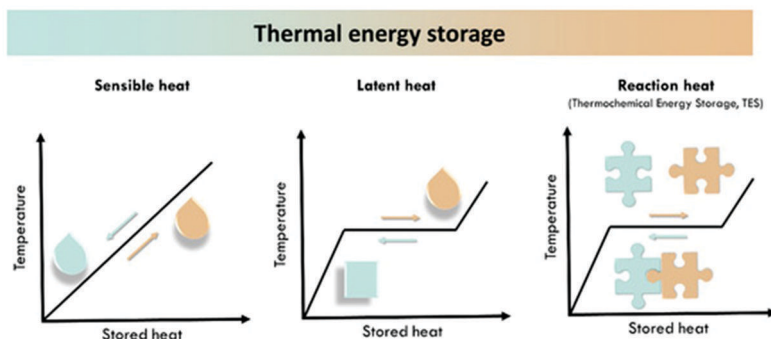


Figure 14. Schematic representation of types of thermal energy storage system. Adapted from reference (66).

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Sensible Heat Storage (SHS)

Sensible heat storage stores thermal energy into the solid or liquid materials by raising its temperature with no change in the phase of the material (67). The amount of heat energy stored into the material depends on the heat capacity of the material, quantity of the material and difference in temperature of the material's initial and final state. Sand, bricks, soil or oil, rock beds and water were used as the storage medium. SHS has various advantages like reduced corrosion, broad operating temperature range and low cost of the material. However, this system exhibits very low energy density. The storage capacity of this was determined by using mass of the medium and specific heat capacity of the medium. It is used in dwellings to supply hot water for offices and houses. Also, used in solar heating systems and air-based systems (68).

Latent Heat Storage (LHS)

Latent heat storage involves the storage of energy during the changes in phase. Materials that were used in LHS are called as phase-change materials (PCMs). The capacity of LHS was determined by the density and enthalpy at the phase transition. The conversion of phases from solid to liquid or liquid to solid takes place in PCM, while charging or discharging the LHS. When the temperature rises, the thermal energy will be absorbed by the PCM and at certain temperature, it starts to melt and involves phase change. Paraffin and salt hydrates have been used in LHS system (11, 68, 69).

Thermochemical Energy Storage

Thermochemical energy storage system involves the dissociation or breaking of bonds and the energy storage takes place during this process. The release of energy occurs during the reverse process. Like other system, the charging, discharging and storing process takes place in this system. This system exhibits higher energy density and more compact in nature. Metal hydrides like magnesium and calcium hydrides, oxides, lead carbonates, calcium carbonates and various organic materials were used for this energy storage system (2). Pumped thermal energy storage system is an example of the above system, which involves the storage of electrical energy as thermal energy. Figure 15 illustrates the charging and discharging process involves in the PTES system. When this system is charging, heat energy will be moved from low temperature reservoir to high-temperature reservoir with the help of heat pump using electrical energy. On discharging, thermal energy will be converted back to the electrical energy by heat engine, which is powered by thermal reservoirs. It has various advantages like high capacity, quick response time and life span upto 30 years (2, 30).

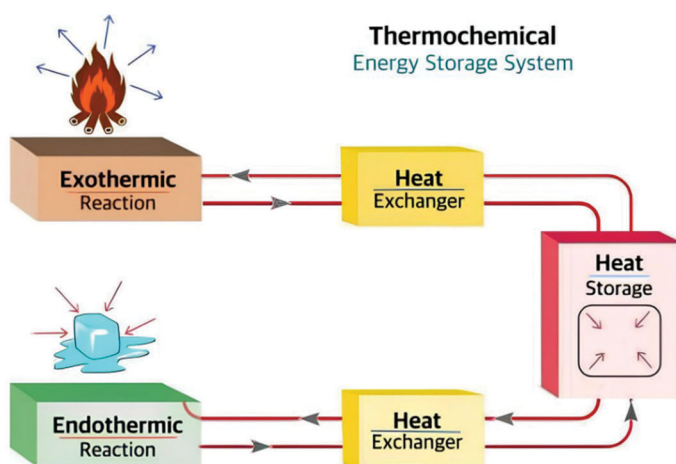


Figure 15. Schematic representation of types of thermochemical energy storage system.

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These TES systems play a crucial role in converting heat energy into electrical energy and storing electricity. Pumped Thermal Energy Storage (PTES) is an example of thermochemical energy storage, where electrical energy is stored as thermal energy. During charging, heat is moved from a low-temperature reservoir to a high-temperature reservoir using a heat pump, while on discharging, thermal energy is converted back to electrical energy using a heat engine. PTES offers advantages like high capacity, quick response time, and a long lifespan of up to 30 years. These diverse TES systems find applications in various sectors, including solar heating systems, dwellings, and grid energy storage, contributing to energy efficiency and sustainability.

Summary

The increase in energy production as per the current need and environmental concern increases the demand for developing better energy storage systems for storing and future purposes. In present,

various types of energy storage systems are available and are categorized based on their physical form of energy such as thermal, electrical, electrochemical, chemical and mechanical energy storage system. The energy density, storage capacity, efficiency, charge and discharge power and response time of the system decides their applications in short term and long-term storage systems. The cost of developing and storing of energies in various forms decides its feasibility in the large-scale applications. Till date various developments in the energy storage systems have been implemented. But, the increasing demand of energy storage systems and the cost-effective way of achieving the global need is becoming the challenging one. SMES are considered as better option in minimizing the power fluctuations for the energy produced from wind and PV, since, they have advantages like shorter response time and high-power density. Similarly, finding the effective integration of energy storage systems with the renewable energy resources will paves way for the betterment in the quality of the power output as well as in improving the efficiency of power output to meet the global demand. In future, the most efficient and cost-effective energy storage systems that meet all global demands are will be developed by bringing various changes in the parts of these energy storage systems.

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